ZZ-H-466c
March 19, 1968
SUPERSEDING
Fed. Spec. ZZ-H-466b
November 8, 1935

FEDERAL SPECIFICATION

HOSE ASSEMBLY, METAL (RUBBER COVERED), HYDROCARBON CONVEYING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a hose with a metal liner which is covered in rubber or synthetic rubber and encased in a woven cotton jacket and assembled with couplings.

1.2 Classification. The hose shall be one type, of the following nominal (inside diameters) sizes: 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, and 4 inches.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specification:

TT-S-735 - Standard Test Fluids, Hydrocarbons.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.
Fed. Std. No. 162 - Hose, Rubber, Visual Inspection Guide for.
Fed. Test Method Std. No. 601 - Rubber Sampling and Testing.

FSC 4720

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, and Standards, at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Specifications:

MIL-H-775 - Hose, Rubber or Fabric (Including Tubing), and Fittings, Nozzles and Strainers, Packaging of.
MIL-C-38404 - Coupling, Hose, Reattachable Screw-On.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-190 - Identification Marking of Rubber Products.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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National Bureau of Standards (NBS) Handbook:

H 28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

United States of America Standards Institute (USASI) Standard:

B2.4 - Hose Couplings Screw Threads.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th St., New York, N.Y. 10016.)

Consolidated Classification Committee

Uniform Freight Classification.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Ill. 60606.)

National Classification Board

National Motor Freight Classification

(Application for copies should be addressed to the National Classification Board, 1616 P St., N.W., Washington, D. C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. Except as otherwise specified herein, the hose and its parts shall be a standard product of the manufacturer or his suppliers. The items furnished shall be new and unused.

3.2 Construction. The hose shall consist of a flexible metal inner lining encased in a tight-fitting rubber or synthetic-rubber tube. A close woven, cotton jacket shall fit tightly over the rubber tube (see fig. 1). The ends of the hose shall be fitted either with male or female or male and female couplings as specified (see 6.2). When female couplings are specified they shall be fitted with leather washers.

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3.3 Materials.

3.3.1 Flexible metal liner. The metal shall be made from a continuous strip of smoothly galvanized steel or tinned bronze.

3.3.2 Rubber tube. The thickness of the rubber or synthetic tube wall, at any point along the metal liner, shall be no less than 1/16 inch (see 4.4.6.1). The rubber or synthetic rubber shall be compounded to seal the flexible metal liner and shall have the physical properties shown in Table I.

TABLE I. Physical properties of rubber and synthetic rubber compounds

Tensile strength, p.s.i. (minimum)	700
Ultimate elongation, percent	
Minimum	200
Maximum	600
Solvent resistance, soluble nonvolatile matter (maximum) percent by wt.	6

3.3.3 Cotton jacket. The cotton jacket shall be a woven fabric, made in accordance with good manufacturing practice, and free from unsightly defects, dirty knots, or lumps. The acceptable quality levels (AQLs) of section 4 shall apply.

3.4 Couplings. Unless otherwise specified (see 6.2), couplings shall conform to class 1, type II of MIL-C-38404 or the coupling shown in figure 2. The composition of the couplings shown in figure 2 shall meet the requirements of class 1 couplings of MIL-C-38404. The couplings shall be free from porosity or other injurious defects. The installed couplings shall not permit the hose to leak or move from its position on the hose when tested in accordance with 4.4.1.1.

3.4.1 Threads. Dimensions and tolerances for coupling threads shall be in accordance with USASI B2.4 and Handbook H28 as applicable and as specified (see 6.2, 6.3, and 6.4).

3.5 Length and size dimensions.

3.5.1 Length. As specified (see 6.2), hose shall be furnished with couplings attached in lengths of 10, 12, 15, 20, or 25 feet between coupling mating faces with a tolerance of $\pm$ 2 percent when examined in accordance with 4.3.1.

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3.5.2 Size. The outside diameter of hose shall fit properly into the couplings. When examined in accordance with 4.3.2, the nominal diameter (i.d.) specified in 1.2 shall have a plus or minus tolerance of 8 percent.

3.6 Physical requirements.

3.6.1 Bending of the flexible metal liner. The flexible metal liner shall be subjected to the bending test specified in 4.4.3. No damage shall result to the liner.

3.6.2 Bending of the complete hose assembly. When subjected to the bending test specified in 4.4.2, the complete hose assembly shall evidence no sign of damage.

3.6.3 Crushing of complete hose assembly. When tested as specified in 4.4.4 the hose shall have no indentation greater than 0.5 inch.

3.6.4 Hydrostatic and proof pressure requirement. Hose assembly shall not leak when subjected to the pressure specified in table II in accordance with paragraph 4.4.1 and 4.4.1.1.

TABLE II. Test loads and pressures

Test	Size of hose, I.D. inches					
	1	1¼	1½	2	2½	3, 3½, 4
Bending of flexible metal liner, applied load in lbs. (min.) (see 4.4.3)	175	250	375	425	450	500
Bending of complete hose, applied load in lbs. (min.)(see 4.4.2)	200	300	350	400	450	500
Crushing of complete hose, applied load in lbs. (min.) (see 4.4.4)	550	800	900	1000	1000	1000
Proof pressure test, p.s.i. (see 4.4.1)	75	75	75	75	75	75
Strength pressure (see 4.4.1.1). Hydrostatic test.						
Low p.s.i.	250	200	150	150	100	100
High p.s.i.	350	350	300	300	250	250

3.6.5 Electrical conductivity. A connection having low resistance shall be provided between couplings in order to ensure the complete elimination of static electricity.

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3.7 Marking. Each length of hose shall be marked in accordance with MIL-STD-190.

3.8 Workmanship. The quality of workmanship shall conform to Fed. Std. No. 162 and the AQL of section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 Lot. For the purpose of sampling, a lot shall consist of all hose of the same size and length offered to the Government at one time.

4.2.2 Unit of product. The unit of product for the examination and tests shall be one length of hose as specified in the contract or order.

4.2.3 Sampling for visual examination. A random sample of hose lengths for visual examination shall be selected from each lot in accordance with MIL-STD-105 at inspection level II and be subject to the examination specified in 4.3. In terms of defects per 100 units, the acceptable quality level shall be 2.5 percent for major defects and 4.0 percent defective for minor defects.

4.2.4 Sampling for tests.

4.2.4.1 Proof pressure test. Each length of hose assembly in the lot shall be subjected to the proof pressure test specified in 4.4.1.

4.2.4.2 Bending and crushing test. Unless otherwise specified (see 6.2), one sample length of hose shall be submitted for each 100 lengths of hose or less. The length of the sample shall be 3 feet for hoses 1 to 1-1/2 inches in diameter, and 4 feet for hoses 2 to 4 inches in diameter.

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4.2.4.3 Sampling of couplings for chemical composition. A minimum 2-ounce sample of millings or dippings shall be taken from each lot for the test specified in 4.4.5. The sample shall be free of dirt, grit, and other foreign matter. The millings or dippings may be taken from the sample used for the bending and crushing test specified in 4.4.3 and 4.4.4.

4.3 Examination. Hose sampled in accordance with 4.2.3 shall be visually examined to determine conformance with the requirements of this specification. The classification of defects in Fed. Std. No. 162 shall be used to determine and evaluate defects. In addition to the defects listed in Fed. Std. 162 hose shall be examined for the following:

Major:

101	Couplings not as specified.
102	Couplings loose.
103	Length of hose not as specified.
104	Threads not as specified.
105	Swivel on female fitting does not turn freely.
106	Inside diameter not within tolerance.

Minor:

201	Marking not as specified.

4.3.1 Length. The hose length, selected in accordance with 4.2.3, shall be determined by method 2411 of Fed. Std. No. 601.

4.3.2 Inside diameter. Prior to performing the bending tests specified in 4.4.2 and 4.4.3, hose samples selected in accordance with 4.2.4.2 shall be used to determine whether the inside diameter requirements of 3.5.2 have been met. The inside diameter shall be determined by inside micrometers or spring calipers and checked with a micrometer caliper at diameters approximately 180° apart in approximately the same plane on two samples.

4.4 Quality conformance tests.

4.4.1 Proof test, air pressure. Each length of hose shall be subjected to an internal air pressure of 75 p.s.i. and held at this pressure for 5 minutes. If a pressure drop of more than 5 p.s.i. develops, the hose length shall be retested by immersion in water, and while the hose is under 75 p.s.i. pressure, examining it for signs of leakage. Any leakage shall be cause for rejection of the hose.

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4.4.1.1 Hydrostatic test. From the hoses tested in 4.4.1, one hose length shall be selected and filled with water. While in a straight position it shall be subjected to the lower hydrostatic pressure shown in table II for 5 minutes; no leakage shall occur. The pressure shall then be increased to the higher hydrostatic value at the rate of 100 p.s.i. per minute. No leakage shall occur while the hose is held at this pressure for 1 minute.

4.4.1.2 Electrical conductivity. While being subjected to the air pressure test specified in 4.4.1, each length of dry hose shall be tested in a 6 volt d.c. electrical circuit containing either a bell or a bulb for electrical conductivity. The bell shall ring or the bulb shall light when the couplings on each end of the assembled hose is used to complete the circuit.

4.4.2 Bending test for hose. Hose selected in accordance with 4.2.4.2 shall be bent to a U-shape between the flat heads of a compression machine and subjected to the load specified in table II. The hose shall be placed so that the bend will not be directly between the compression heads but subjected to the load transmitted through the legs of the U formed by the hose. The legs shall not withhold any of the load from the bend by coming into contact. The hose shall then be subjected to the test in 4.4.1. Damage to the rubber cover or liner or leakage caused by these tests shall be considered a critical defect.

4.4.3 Bending test for metal liner. Remove the rubber tubing and cotton jacket from the hose tested in 4.4.2. The rubber tubing shall be used in the test specified in 4.4.6. The metal liner shall then be bent U-shaped and tested as specified in 4.4.2. Any deformation of the liner shall be cause for rejection of the lot.

4.4.4 Crushing test. Hose selected in accordance with 4.2.4.2 shall be subjected to the crushing load specified in table II. The load shall be applied to a 1-inch long flat metal block placed longitudinally on top of the hose at three separate points. Any indentation greater than 0.5 inch shall be cause for rejection of the lot.

4.4.5 Chemical composition of couplings. Samples selected in accordance with 4.2.4.3 shall be analyzed in accordance with method 111.1 or 112.1 of Fed. Test Method Std. No. 151 to determine conformance with 3.4. The Government will accept laboratory verification or certificate compliance from the coupling vendor stating that the couplings meet the requirement of 3.4.

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4.4.6 Tensile strength and elongation of rubber tube. The rubber tubing removed from the hose samples used in the bending test (see 4.4.3) shall be tested in accordance with methods 4111 and 4121 of Fed. Test Method Std. No. 601 to determine if the tensile strength and elongation requirements have been met.

4.4.6.1 Thickness of rubber tube wall. Prior to determination of the tensile strength, the rubber thickness shall be measured in accordance with method 2021 of Fed. Test Method Std. No. 601 except that a 1/4 inch ball shall be attached to the face of the anvil of the dial indicator. The thickness of the rubber, measured from the part of the rubber in contact with the metal liner to the outer surface of the rubber revealed when the cotton jacket has been buffed away, shall conform to the thickness specified in 3.3.2.

4.4.7 Solvent resistance of rubber tubing. The rubber tubing shall be extracted in accordance with method 6511 of Federal Test Method Std. No. 601. The immersion fluid shall be type II of TT-S-735.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the hose shall be inspected to determine conformance with the applicable requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The hose shall be preserved and packaged in accordance with the applicable level A requirements of MIL-H-775.

5.1.2 Level C. The hose shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The hose shall be packed in accordance with the applicable level A requirements of MIL-H-775.

5.2.2 Level B. The hose shall be packed in accordance with the applicable level B requirements of MIL-H-775.

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5.2.3 Level C. The complete equipment shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, the hose shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, the hose shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The hose covered by this specification is intended for handling aliphatic and aromatic hydrocarbons. The metal liner assures positive dissipation of static.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Size of hose and length of hose required (see 1.2 and 3.5).

(c) The kind of couplings required (see 3.2 and 3.4).

(d) Type of threads for couplings required (see 3.4.1).

(e) If the sampling requirements for bend and crush tests are different (see 4.2.4.2).

(f) Level of preservation and level of packing required (see 5.1 and 5.2).

(g) Marking required (see 5.3).

6.3 International interest. Certain provisions pertaining to the threading of couplings for nominal hose sizes over two inches, covered by this specification (see 3.4.1 and 6.4), are the subject of International Standardization Agreement, NATO STANAG No. 2761. When

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amendment, revision or cancellation of this specification is proposed, the departmental custodians will inform their respective departmental standardization offices so that appropriate action may be taken respecting the international agreement concerned.

6.4 Types of thread. Types 1, 2, 3, and 4 thread designations, shown in table III, are referenced to provide a means of comparison with the thread designations specified in ZZ-H-466b. These thread designations are referenced in international standardization documents but are not listed by type 1, 2, 3, and 4 in USASI B2.4 or in Handbook H28.

TABLE III. Types of threads

Type of thread	Nominal size inches	Threads per inch	Thread designation and symbol in USASI B2.4 and Handbook H28
1	1	11 1/2	NPSH 1/
1	1 1/4	11 1/2	NPSH
1	1 1/2	11 1/2	NPSH
1	2	11 1/2	NPSH
2	2 1/2	8	NPSH
2	3	8	NPSH
2	3 1/2	8	NPSH
2	4	8	NPSH
3	2 1/2	7 1/2	NH 2/
3	3	6	NH
3	3 1/2	6	NH
4	4	6	NH(SPL) 3/

1/ American National Hose coupling thread.
2/ American National fire-hose coupling.
3/ American National fire-hose coupling (special).

MILITARY INTEREST:
Custodians:
Army - ME
Navy - YD
Air Force - 82

User activity:
Navy - MC

Preparing activity:
Navy - YD

CIVIL AGENCY INTEREST:
GSA - FSS

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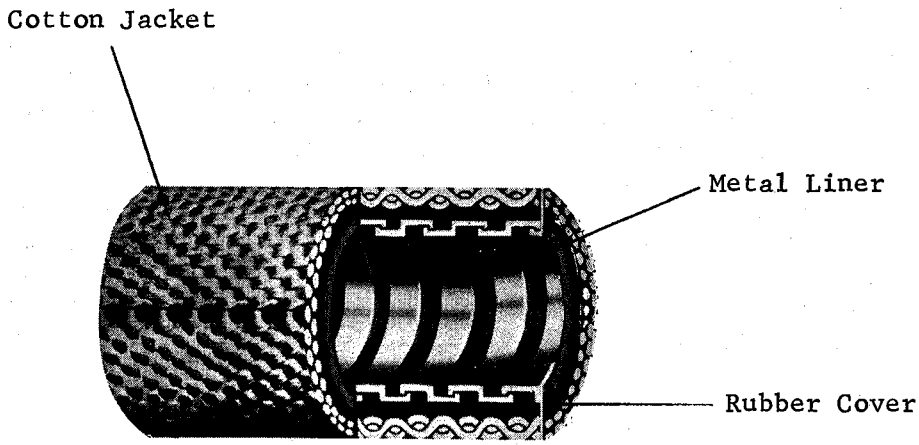
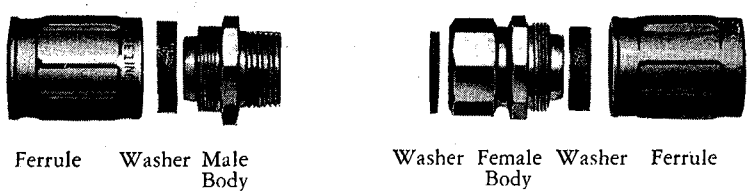


Figure 1. Metal Lined Hose
(for illustrative purposes only)



Male Female

Figure 2. Hose couplings

Order for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ZZ-H-466c - Hose Assembly, Metal (Rubber Covered), Hydrocarbon Conveying

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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